
IAL Biology Unit 2

WBI12

Cells, Development, Biodiversity and Conservation

A* revision and exam-technique guide

Built from analysis of 21 past papers and mark schemes,
May 2019 to May 2026

WBI12/01 · 80 marks · 1 hour 30 minutes
40% of the IAS · 20% of the full IAL

1 • What the papers actually look like

Across all 21 papers: **80 marks, 90 minutes, usually eight questions** (occasionally six, seven or nine). Multiple-choice items appear embedded within questions rather than as a separate section. The structure is the same as Unit 1, and so is the mark ramp.

Question	Mean marks	Range	Implication
Q1	6.0	5–9	Short and usually single-topic. Clear these quickly.
Q2	6.8	5–9	
Q3	7.8	5–12	
Q4	9.8	8–12	Multi-part and data-heavy. Q5-Q8 carry roughly 60% of the paper. Protect your time for them.
Q5	10.4	7–13	
Q6	11.4	8–14	
Q7	12.7	11–15	
Q8	14.3	11–16	

Command words, counted across all 21 papers

Command	Per paper	What it requires
Explain	7.4	Dominant. Needs a reason — <i>because, so, therefore</i> . A description scores nothing here.
Give / State / Name	9.1 combined	Unusually high in this unit — much of Topic 3 and 4 is recall of structures, functions and definitions. Answer in a word or phrase; do not explain.
Calculate	3.8	Every paper. Magnification is the single most common type.
Describe	3.7	Say what happens or what the data shows, with figures and units. No reason needed.
Suggest	2.9	Unfamiliar context by design. Never leave blank.
Comment / Discuss / Evaluate	1.8 combined	Needs both sides and an explicit judgement.
Compare / Draw / Label	1.5 combined	Compare must address both items per sentence. Drawing questions have their own conventions (§4.6).

Topic frequency across the 21 papers

Topic area	Papers (of 21)	Verdict
Maths and practical skills	21	Guaranteed — every single paper. If you prepare nothing else, prepare these.
Cell ultrastructure and organelles	21	
Gametes and fertilisation	21	
Prokaryotes vs eukaryotes	19	Near-certain. Nine papers in ten.
Microscopy, magnification and resolution	19	
Mitosis and the cell cycle	19	
Plant cell structure and the cell wall	19	
Biodiversity and the index of diversity	16	
Plant fibres and sustainability	16	Very likely. Two papers in three.
Niche and adaptation	14	
Starch vs cellulose comparison	13	
Conservation, zoos and seed banks	13	
Phenotype, genotype and variation	11	
Classification and molecular phylogeny	11	Common. Roughly half of papers.
Protein secretion pathway	10	
Stem cells and differentiation	10	
Minerals and plant nutrition	10	
Meiosis and sources of variation	9	
Gene expression and epigenetics	6	Periodic. Do not skip — crossing over appeared 34 times across the mark schemes, more than most "common" topics.
Drug testing and clinical trials	4	
Antimicrobials from plants	4	
Natural selection, evolution and speciation	1	

TWO FINDINGS WORTH ACTING ON

Evolution is not where you think it is. Across 21 papers, "natural selection" appeared in two, "speciation" in one, and "evolution" in none. **Adaptation and niche**, by contrast, appeared in 14–17. Unit 2 tests how organisms *are* adapted and how diversity *is measured*, not the mechanism of evolutionary change — that is developed in Unit 4. Revise adaptation and the diversity index thoroughly; treat selection theory as background.

This unit rewards recall more than Unit 1 does. "Give", "State" and "Name" together run to over nine marks per paper. Organelle functions, gamete structures, mineral roles and definitions are straightforward marks that are lost only through not having learned them.

2.1 Cell ultrastructure

In all 21 papers. Learn each organelle as **structure → function**, because questions ask for both and frequently give you one and ask for the other.

Organelle	Structure	Function
Nucleus	Double membrane (nuclear envelope) with pores; contains chromatin and a nucleolus	Holds DNA; site of transcription; the nucleolus makes ribosomes
Mitochondrion	Double membrane; inner folded into cristae ; fluid matrix	Aerobic respiration, producing ATP . Cristae give a large surface area.
Rough ER	Membrane sheets studded with ribosomes	Synthesis and transport of proteins
Smooth ER	Membrane tubules, no ribosomes	Synthesis of lipids and steroids
Golgi apparatus	Stack of flattened membrane sacs with vesicles	Modifies, processes and packages proteins; makes lysosomes
Lysosome	Membrane-bound vesicle of hydrolytic enzymes	Digests worn-out organelles and engulfed material
Ribosome	80S in eukaryotes, 70S in prokaryotes; no membrane	Site of translation / protein synthesis
Centriole	Two bundles of microtubules at right angles	Forms the spindle in cell division
Chloroplast (plant)	Double membrane; thylakoids stacked into grana; stroma	Photosynthesis
Cell wall (plant)	Cellulose microfibrils	Support; prevents bursting; freely permeable
Vacuole (plant)	Large, surrounded by the tonoplast	Stores cell sap; maintains turgor

THE PROTEIN SECRETION PATHWAY — A RECURRING 4-MARK SEQUENCE

Nucleus (gene transcribed to mRNA, leaves through a nuclear pore) → **ribosome on the rough ER** (protein synthesised) → transported through the rER → **vesicle** buds off → **Golgi apparatus** (protein **modified**, e.g. by adding carbohydrate) → **secretory vesicle** → fuses with the **cell surface membrane** → released by **exocytosis**.

Where the question gives a cell type, link it: a cell secreting large amounts of protein will have **many mitochondria** (ATP for transport and exocytosis), **extensive rough ER** and a **prominent Golgi**.

2.2 Prokaryotic vs eukaryotic cells

Feature	Prokaryotic	Eukaryotic
Nucleus	No nucleus ; circular DNA free in the cytoplasm, not associated with proteins	True nucleus with a double membrane; linear DNA associated with histone proteins
Membrane-bound organelles	Absent	Present
Ribosomes	70S (smaller)	80S (larger)
Cell wall	Present, made of peptidoglycan (murein)	Absent in animals; cellulose in plants
Plasmids	Often present	Absent
Size	Typically 0.5–5 µm	Typically 10–100 µm
Other	May have a capsule, flagellum, pili; mesosomes	May have cilia, flagella, centrioles

2.3 Microscopy

THE TWO TERMS EXAMINERS TEST THE DIFFERENCE BETWEEN

Magnification — how many times larger the image is than the actual object.

Resolution — the **minimum distance between two points** at which they can still be distinguished as separate. Resolution is limited by the **wavelength** of the radiation used.

Electron microscopes have far higher resolution because **electrons have a much shorter wavelength than light**. That sentence is the mark.

	Light microscope	TEM	SEM
Resolution	About 200 nm	About 0.1 nm	About 3–10 nm
Specimen	Living or dead; colour possible	Must be dead, in a vacuum; very thin section	Must be dead; surface coated
Image	2D, colour	2D, black and white	3D surface , black and white
Limitation	Low resolution	Cannot view living specimens; artefacts may be introduced in preparation	Lower resolution than TEM

MAGNIFICATION — THE MOST COMMON CALCULATION IN THE UNIT (5 PAPERS ASKED FOR IT DIRECTLY)

Magnification = image size ÷ actual size

Method: measure the image **in mm with a ruler**, convert to the same unit as the actual size, then divide. Units: **1 mm = 1000 µm = 1 000 000 nm**.

Converting before dividing — not after — removes most of the errors. When a **scale bar** is given, measure the bar and divide its measured length by the length it represents.

2.4 Gametes and fertilisation

In all 21 papers, and "acrosome" appeared 25 times across the mark schemes.

Sperm cell — adaptation	Function
Acrosome	Contains hydrolytic (digestive) enzymes to digest through the zona pellucida
Many mitochondria	Produce ATP for movement of the flagellum
Flagellum / tail	Propels the sperm towards the egg
Haploid nucleus	So that fusion restores the diploid number

Small, streamlined, little cytoplasm	Reduces resistance, aiding movement
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Egg cell — adaptation	Function
Zona pellucida	Glycoprotein layer; becomes impenetrable after fertilisation
Cortical granules	Release contents to thicken the zona pellucida — prevents polyspermy
Lipid droplets / food reserves	Supply energy and materials to the developing embryo
Follicle cells	Protective outer layer
Haploid nucleus , large cell	Restores diploid number on fusion; large volume supports early development

FERTILISATION — THE FIVE-STEP SEQUENCE

- Sperm reaches the egg and binds to the zona pellucida.
- Acrosome reaction:** the acrosome releases **hydrolytic enzymes** which **digest through the zona pellucida**.
- The sperm **cell membrane fuses** with the egg cell membrane; the sperm nucleus enters.
- Cortical reaction:** cortical granules release their contents into the zona pellucida, which **thickens and hardens, preventing polyspermy** — the entry of further sperm.
- The two **haploid nuclei fuse** to form a **diploid zygote**.

2.5 The cell cycle and mitosis

In 19 of 21 papers, and the **mitotic index** calculation appeared in 12.

Stage	What happens
Interphase — G1	Cell grows; organelles replicate; proteins synthesised
Interphase — S	DNA replicates , so each chromosome becomes two sister chromatids
Interphase — G2	Further growth; the cell prepares for division
Prophase	Chromosomes condense and become visible; nuclear envelope breaks down ; centrioles move to opposite poles and the spindle forms
Metaphase	Chromosomes line up at the equator , attached to spindle fibres by their centromeres
Anaphase	Centromeres divide ; sister chromatids are pulled to opposite poles as the spindle fibres shorten
Telophase	Chromosomes decondense; nuclear envelopes re-form around each set
Cytokinesis	The cytoplasm divides, giving two genetically identical diploid cells

MITOTIC INDEX

Mitotic index = number of cells in mitosis ÷ total number of cells (× 100 for a percentage)

Count only cells in prophase, metaphase, anaphase or telophase — cells in interphase are **not** in mitosis. A high mitotic index indicates rapid cell division, as in a growing tissue, a root tip, or a tumour.

Purposes of mitosis: growth, repair and replacement of cells, and asexual reproduction. It produces cells that are **genetically identical** to the parent cell.

2.6 Meiosis and the sources of variation

In only 9 of 21 papers, but "crossing over" appeared **34 times** across the mark schemes — more than most higher-frequency topics. When it comes up, it carries marks.

Meiosis involves **two divisions** and produces **four haploid, genetically different** cells from one diploid cell.

THREE SOURCES OF GENETIC VARIATION — NAME ALL THREE

- Crossing over** — in **prophase I**, homologous chromosomes pair up as **bivalents**, **chiasmata** form, and **sections of chromatid are exchanged**, producing new combinations of alleles.
- Independent assortment** — in **metaphase I**, the homologous pairs line up at the equator in a **random orientation**, so which chromosome of each pair goes to which pole is random. The same occurs for chromatids in metaphase II.
- Random fertilisation** — any sperm may fuse with any egg.

	Mitosis	Meiosis
Divisions	One	Two
Daughter cells	Two	Four
Chromosome number	Diploid, unchanged	Halved — diploid to haploid
Genetic identity	Identical to parent	Genetically different
Homologous pairing	Does not occur	Occurs in prophase I, with crossing over
Purpose	Growth, repair, asexual reproduction	Production of gametes

2.7 Stem cells and differentiation

Type	Can become	Source
Totipotent	Any cell type, including placental tissue	Zygote and the cells of the very early embryo
Pluripotent	Any cell type of the body , but not placental tissue	Inner cell mass of the blastocyst — embryonic stem cells
Multipotent	A limited range of cell types	Adult tissues, e.g. bone marrow

Why cells differ despite identical DNA: all cells of an organism contain the **same genes**, but in each cell type **different genes are expressed** — switched on or off. The proteins made therefore differ, and these determine the cell's structure and function.

Control of gene expression is by **transcription factors**, proteins that bind to a specific region of DNA near a gene and switch transcription on or off. Epigenetic control operates through **DNA methylation**, which switches genes off, and **histone acetylation**, which makes DNA more accessible and switches genes on.

THE STEM CELL ETHICS QUESTION — STRUCTURE THE ANSWER

For: potential to treat conditions currently untreatable — Parkinson's, spinal injury, type 1 diabetes; embryos used are often surplus from IVF and would otherwise be destroyed.

Against: the embryo is destroyed; disagreement about the moral status of an embryo and when life begins; questions of consent; risk of tumour formation in treatment.

Then judge. Note that adult and induced pluripotent stem cells avoid much of the objection. An answer that lists both sides without reaching a judgement does not reach the top band.

2.8 Phenotype, genotype and variation

Phenotype = genotype + environment.

	Continuous variation	Discontinuous variation
Control	Polygenic — many genes	Monogenic — one or very few genes
Environment	Strongly influenced	Little or no influence
Range	A continuous range of values	Distinct, separate categories
Examples	Height, mass, skin colour	Blood group, ability to roll the tongue
Graph	Normal distribution curve; histogram	Bar chart

3 · Topic 4 — Biodiversity and Natural Resources

3.1 Biodiversity and how it is measured

- **Species richness** — the **number of different species** in a habitat. Simple, but it ignores how common each is.
- **Index of diversity** — takes account of both the number of species **and** the number of individuals of each, so it is the better measure.
- **Genetic diversity** — the variety of alleles within the gene pool of a species or population; measured by the **heterozygosity index**.

INDEX OF DIVERSITY — THE FORMULA IS GIVEN IN THE PAPER, SO MARKS ARE FOR USING IT

$$D = \frac{N(N - 1)}{\sum n(n - 1)}$$

N = total number of organisms of **all** species. **n** = number of individuals of **each** species.

Method: compute n(n–1) for every species, add them to get the denominator, then divide. **A higher D means higher diversity.**

The common errors: using the number of species instead of the number of individuals for N; forgetting to sum across all species; and not stating what the value means when asked to comment.

Heterozygosity index = **number of heterozygotes** + **total number of individuals in the population**. A higher value means greater genetic diversity, which matters because a population with low genetic diversity is less able to adapt to environmental change and is more vulnerable to disease.

3.2 Niche and adaptation

DEFINITION

A **niche** is the **role and position of a species within its environment**, including all its interactions with **biotic and abiotic** factors and how it obtains its resources.

Two species cannot occupy the **same niche in the same habitat** — they would compete, and one would be out-competed.

Type of adaptation	What it is	Example
Anatomical	A structural feature of the body	Thick fur; a cactus's spines and shallow spreading roots; a streamlined body
Physiological	An internal process or biochemical feature	Producing concentrated urine; hibernating metabolic changes; CAM photosynthesis
Behavioural	A way the organism acts	Migration; nocturnal activity; basking in the sun

Questions nearly always present an unfamiliar organism and ask you to **suggest** how a named feature is an adaptation. Always answer in the form "**feature → how it works → benefit to survival**". Naming the feature alone scores nothing.

3.3 Classification and molecular phylogeny

The taxonomic hierarchy: **Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species**. The **binomial name** is Genus (capitalised) then species (lower case), both italicised or underlined.

The **three domains** are **Bacteria, Archaea and Eukarya**, distinguished principally by differences in **ribosomal RNA sequences**, as well as in cell wall composition and membrane lipids.

MOLECULAR EVIDENCE FOR RELATEDNESS — THE THREE METHODS

1. Comparing **DNA base sequences**.
2. Comparing the **amino acid sequences of proteins** such as cytochrome c or haemoglobin.
3. **Immunological** comparison of proteins — the greater the precipitate formed, the more similar the proteins.

The principle in all three: **the more similar the sequences, the more closely related the species, and the more recent their common ancestor**. This evidence is more reliable than physical appearance, because similar-looking organisms may be unrelated and have converged.

3.4 Plant cell structure, cellulose and fibres

In 19 of 21 papers. "Cellulose" appeared 76 times across the mark schemes and "microfibril" 40 — the two highest-frequency content terms in the whole unit.

Structure	Function
Cell wall	Made of cellulose microfibrils ; provides support and prevents the cell bursting; freely permeable
Middle lamella	Layer of calcium pectate between adjacent cells, sticking them together
Plasmodesmata	Cytoplasmic channels through the cell wall, allowing transport and communication between cells
Pits	Thin regions of the wall allowing water transport
Amyloplast	Stores starch
Vacuole / tonoplast	Stores cell sap; maintains turgor pressure

WHY CELLULOSE IS STRONG — THE FULL ANSWER

Cellulose is made of **β-glucose** monomers joined by **1,4 glycosidic bonds**. Alternate molecules are **rotated through 180°**, which allows the chain to be **long and straight**. **Hydrogen bonds** form **between parallel chains**, holding them together in bundles called **microfibrils**, which group into **macrofibrils**. The large number of hydrogen bonds gives cellulose very high **tensile strength**.

	Starch	Cellulose
Monomer	α-glucose	β-glucose
Chain shape	Coiled (amylose) or branched (amylopectin)	Long, straight, unbranched
Bonding between chains	None significant	Many hydrogen bonds → microfibrils
Function	Energy storage — compact, rapidly hydrolysed	Structural — high tensile strength

PLANT FIBRES AND SUSTAINABILITY

- **Sclerenchyma fibres** — long cells with thick walls containing **lignin**; dead at maturity; provide support and high tensile strength.
 - **Xylem vessels** — hollow tubes of dead cells with **lignified** walls; transport water and also provide support.
 - **Advantages of plant fibres over oil-based materials: renewable** and sustainable; **biodegradable**; require **less energy** to produce, so lower carbon dioxide emissions; often cheaper.
 - **Disadvantages:** more variable in quality; generally weaker than some synthetics; growing them may compete with food production for land.
- Starch and nitrate practicals** aside, the recurring practical here is testing the **tensile strength** of plant fibres by hanging increasing masses until the fibre breaks, controlling fibre length, diameter and the way the mass is added.

3.5 Minerals and plant nutrition

Mineral	Needed for	Deficiency symptom
Nitrate	Amino acids and proteins, nucleic acids, chlorophyll	Stunted growth; yellowing of older leaves
Magnesium	A component of chlorophyll	Chlorosis — yellowing of leaves; reduced photosynthesis
Calcium	Calcium pectate in the middle lamella of the cell wall	Stunted growth; poor root development; young leaves distorted

3.6 Drug testing and clinical trials

In only 4 of 21 papers, but "placebo" appeared 22 times in the mark schemes — when this topic appears it carries several marks at once.

William Withering used a **digitalis soup** made from foxglove to treat dropsy, varying the proportion of foxglove to find a dose that was effective without being toxic. By modern standards his method lacked a control group, a placebo, randomisation and any blinding.

Stage	Tested on	Purpose
Preclinical	Cells, tissues, then animals	Toxicity and basic efficacy
Phase I	A small number of healthy volunteers	Safety, dosage, side effects
Phase II	A small group of patients	Whether the drug actually works
Phase III	A large group of patients	Comparison with an existing treatment or placebo; detection of rarer side effects

THE THREE TERMS, DEFINED PRECISELY

Placebo — a substance with **no active ingredient**, given so that any effect of the drug can be distinguished from the psychological effect of being treated.

Double-blind — **neither the patient nor the doctor** knows who has received the drug and who the placebo, which removes bias from both the patient's reporting and the doctor's assessment.

Randomised — patients are allocated to groups at random, so the groups are comparable and allocation bias is avoided.

Ethical tension to note in an evaluation: using a placebo means some patients knowingly go untreated, which is hard to justify for a serious condition where an effective treatment already exists.

3.7 Antimicrobials from plants

The standard practical: discs soaked in plant extracts are placed on an agar plate inoculated with bacteria, and the **diameter of the zone of inhibition** — the clear area where bacteria have not grown — is measured after incubation. A larger zone indicates a more effective antimicrobial.

Aseptic technique points that earn marks: sterilise the inoculating loop by flaming; work near a Bunsen flame; lift the lid of the Petri dish only slightly and at an angle; seal the dish with tape but do not seal it completely, so that anaerobic pathogens cannot grow; incubate at no more than 25 °C in a school laboratory. Controls: a disc soaked in the **solvent alone**, and a disc with a known antibiotic.

3.8 Conservation — zoos and seed banks

	Methods and conditions	Problems and limitations
Seed banks	Seeds are dried to a low moisture content and stored at low temperature (typically −20 °C), which slows metabolism. Viability is tested periodically by germinating a sample.	Not all species' seeds survive drying and freezing (recalcitrant seeds); samples must be periodically regrown, which is costly; genetic diversity may be lost over successive regenerations; a stored seed does not conserve the habitat.
Zoos	Captive breeding programmes ; studbooks record ancestry so that related individuals are not bred together; assisted reproduction such as IVF and embryo transfer; reintroduction to the wild; research and education.	Small gene pool leading to inbreeding and loss of genetic diversity; captive-bred animals may not learn natural behaviours such as hunting or avoiding predators; disease spreads readily; expensive; reintroduction often fails if the original habitat is still degraded; ethical objections to confinement.

In situ conservation protects species in their natural habitat — nature reserves, national parks, marine protected areas. **Ex situ** conservation is outside it — zoos, botanic gardens, seed banks. In situ preserves the habitat and the species' natural behaviour and is generally preferred; ex situ is a safeguard when in situ protection is not feasible.

4 · Skills the papers test every time

4.1 The calculations that actually come up

All 21 papers contained calculations, averaging 3.8 per paper. Taken from the actual papers, in order of how often they appeared:

Type	Method	The trap
Magnification (most common)	magnification = image size ÷ actual size. Measure the image in mm, then convert.	Not converting units before dividing. 1 mm = 1000 µm = 1 000 000 nm.
Actual size from a scale bar	Measure the scale bar; divide its measured length by the length it represents to get the magnification; then divide the measured structure by that.	Measuring the bar carelessly — a 1 mm error propagates through everything.
Mitotic index (12 papers)	cells in mitosis ÷ total cells (× 100 for a percentage)	Counting interphase cells as dividing, or dividing by the wrong total.
Index of diversity	D = N(N–1) ÷ Σn(n–1) ; build a column of n(n–1) and sum it.	Using the number of species rather than individuals for N.
Heterozygosity index	heterozygotes ÷ total individuals	Confusing it with the index of diversity.
Volume and surface area	Formulae are given in the question — substitute carefully.	Using diameter where the formula wants radius.
Percentage and percentage change	change ÷ original × 100	Dividing by the final value.
Number of cells after n divisions	2 ⁿ	Confusing the number of divisions with the number of cells.
Standard deviation and error bars	Calculator function. Overlapping error bars suggest the difference is not significant; non-overlapping suggests it may be.	Claiming a difference is significant when the bars overlap.
Correlation coefficient (6 papers)	Interpret the value: closer to +1 or –1 means a stronger correlation.	Saying a strong correlation proves causation.

KNOW WHICH STATISTICAL TEST DOES WHAT — MULTIPLE-CHOICE ITEMS TEST EXACTLY THIS

Index of diversity — compares biodiversity between habitats.

Mitotic index — the proportion of cells actively dividing.

Chi-squared — compares observed with expected results (for categorical data).

Student's t-test — compares the means of two independent groups.

Correlation coefficient — measures the strength of a relationship between two variables.

Standard deviation — measures the spread of data about the mean.

4.2 Scientific drawings

Drawing questions have their own conventions, and the marks are for the conventions as much as the biology:

- Use a **sharp pencil**; draw **clear, continuous, unbroken lines** with no sketching or shading.
- Draw only what you can actually see — **no interpretation** of structures not visible.
- Draw structures in the **correct proportion** to each other.
- **Label lines must be straight, drawn with a ruler, must not cross**, and must touch the structure they label.
- Include a **title** and the **magnification or a scale bar**.

5 · Mark-scheme phrase bank

These were the highest-frequency content terms across the 21 mark schemes. The left column is what scores.

Context	Write this	Not this
Cellulose strength (76 mentions)	" β-glucose , alternate molecules rotated 180° , giving straight chains; hydrogen bonds between parallel chains form microfibrils "	"Cellulose is strong because it is a polymer"
Crossing over (34)	"In prophase I , homologous chromosomes form bivalents, chiasmata form and sections of chromatid are exchanged "	"The chromosomes swap genes"
Acrosome (25)	"Releases hydrolytic enzymes which digest through the zona pellucida "	"The acrosome helps the sperm get in"
Cortical reaction	"Cortical granules release their contents, the zona pellucida thickens, preventing polyspermy "	"It stops other sperm"
Electron microscope	" Electrons have a shorter wavelength than light, giving higher resolution "	"Electron microscopes are more powerful"
Resolution	"The minimum distance between two points at which they can be distinguished as separate"	"How clear the image is"
Independent assortment	"In metaphase I , homologous pairs line up in a random orientation "	"The chromosomes are random"
Why cells differ	"All cells contain the same genes , but different genes are expressed , so different proteins are made"	"They have different DNA"
Transcription factors	"Bind to a specific region of DNA and switch transcription on or off "	"They control the gene"
Index of diversity	"A higher D means higher diversity , taking account of both species number and the number of individuals of each"	"D shows how many species there are"
Niche	"The role and position of a species in its environment, including all interactions with biotic and abiotic factors"	"Where an organism lives"
Any adaptation question	Feature → how it works → benefit to survival	Naming the feature only
Placebo and blinding	"A substance with no active ingredient "; " neither patient nor doctor knows which group a patient is in, which removes bias "	"A fake pill"; "it's a fair test"
Molecular phylogeny	"The more similar the sequences , the more closely related and the more recent the common ancestor "	"They look alike so they are related"
Describing a graph	Quote figures with units and name the trend	"It goes up"

6 • The mistakes that cost the grade

1. Not converting units before a **magnification** calculation — the commonest single error in this unit.
2. Confusing **magnification** with **resolution**.
3. Counting interphase cells when calculating the **mitotic index**.
4. Using the number of **species** instead of **individuals** for N in the diversity index.
5. Calculating an index and then not **saying what it means** when asked to comment.
5. Giving an organelle's name when the question asked for its **function**, or the reverse.
7. Describing the secretion pathway without naming the **Golgi's modifying role**.
8. Saying prokaryotes have "no DNA" rather than **no nucleus**; the DNA is circular and free in the cytoplasm.
9. Forgetting that prokaryotic ribosomes are **70S** and eukaryotic **80S**.
10. Describing fertilisation without the **cortical reaction** and **polyspermy**.
11. Writing that the acrosome "breaks" the zona pellucida rather than releasing **hydrolytic enzymes** that **digest** it.
12. Muddling the mitosis stages — particularly **anaphase**, where it is the **centromeres that divide** and **chromatids** that separate.
13. Saying meiosis produces "two" cells, or forgetting that they are **genetically different**.
14. Naming only one source of variation in meiosis when the question wants both crossing over **and** independent assortment.
15. Confusing **totipotent** with **pluripotent** — the distinction is placental tissue.
16. Saying differentiated cells "have different DNA". They have the same genes, differently expressed.
17. Explaining cellulose strength without mentioning **hydrogen bonds between chains**.
18. Saying cellulose is made of glucose without specifying **β-glucose**.
19. Confusing the **middle lamella** (calcium pectate, between cells) with the cell wall.
20. Giving a plant mineral's deficiency symptom without its **role**, or the reverse.
21. Defining a **niche** as a place rather than a role.
22. Describing an adaptation without stating the **benefit to survival**.
23. In conservation questions, listing methods but not the **problems** when the question asks for an evaluation.
24. Claiming a correlation or a non-overlapping error bar **proves** something.
25. Over-writing on Q1–Q3 and leaving too little time for Q7 and Q8, which hold over a quarter of the paper.

7 • Final checklist

Can you, from memory...	Can you, from memory...
• Give the structure and function of eleven organelles? • Trace a protein from gene to secretion, naming every structure? • Give six differences between prokaryotic and eukaryotic cells? • Define magnification and resolution, and say why electron microscopes resolve better? • Do a magnification calculation with a unit conversion, correctly? • Use a scale bar to find an actual size? • Give five sperm adaptations and five egg adaptations with functions? • Describe fertilisation in five steps including both reactions? • Name the three interphase stages and say what happens in S? • Describe all four mitosis stages with the right events? • Calculate a mitotic index and say what a high value means? • Give three sources of variation in meiosis, with the stage each occurs? • Give six differences between mitosis and meiosis?	• Distinguish totipotent, pluripotent and multipotent with sources? • Explain why differentiated cells differ despite identical DNA? • Argue both sides of embryonic stem cell use and reach a judgement? • Distinguish continuous from discontinuous variation on four criteria? • Use the index of diversity formula and interpret the result? • Define a niche properly? • Classify an adaptation as anatomical, physiological or behavioural? • Give the taxonomic hierarchy and the three domains? • Give three molecular methods of establishing relatedness? • Explain cellulose's tensile strength in four linked steps? • Compare starch and cellulose on four criteria? • Give four advantages and two disadvantages of plant fibres? • Give the role and deficiency symptom for nitrate, magnesium and calcium? • Describe the four stages of drug trials and define placebo, double-blind and randomised? • Give seed bank conditions and three problems; zoo methods and four problems? • State which statistical test is used for what?

THE SHORT VERSION

This unit rewards learned detail more than Unit 1 does — over nine marks a paper come from "give", "state" and "name". Learn the organelle table, the gamete adaptations, the mitosis stages and the mineral roles cold, because those marks are lost only to gaps in recall. Then practise the four calculations until they are automatic, since every paper has them and magnification alone appears in nearly all. Everything else is explanation, and explanation is won by naming the mechanism rather than describing the outcome.